

RESEARCH ARTICLE

Polydopamine-Mediated, Amphiphilic Poly(Carboxybetaine Methacrylamide-*r*-Trifluoroethyl Methacrylate) Coating with Resistance to Marine Diatom Adhesion and Silt Adsorption

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Marine biofouling—the adhesion of marine organisms onto a ship hull—causes increased fuel consumption, leading to massive carbon dioxide emissions. Many attempts are made to address this issue, and antifouling polymer coatings are extensively investigated owing to their environmental friendliness. Zwitterionic polymers, polysaccharides, and polyethylene glycol are frequently used as surface coatings, demonstrating excellent marine antifouling performance. However, these hydrophilic polymer coatings have a major drawback: when exposed to sediment, various minerals are easily adsorbed by the coatings, causing them to lose their inherent antifouling properties. Amphiphilic polymer coatings have therefore been proposed as alternatives to hydrophilic polymer coatings. In this study, the synthesis of amphiphilic copolymers composed of carboxybetaine methacrylamide and trifluoroethyl methacrylate and the immobilization of these copolymers onto solid surfaces are reported. This method utilizes material-independent surface-coating properties and the metal complex-forming ability of polydopamine to immobilize amphiphilic copolymers onto solid surfaces. The resulting surfaces exhibit good antifouling performance against both diatom adhesion and silt adsorption. As this is a facile and substrate-independent method for immobilizing polymers, an expectation exists for it to be an effective platform for the coating of new marine antifouling polymers.

transport to new environments, which can threaten domestic marine ecosystems.^[4] To solve these economic and ecological problems, surface coatings with antifouling, fouling-release properties, or both have been extensively investigated.^[1] A recent strategy to reduce surface fouling is to coat solid surfaces with hydrophilic polymers.^[5,6] These hydrophilic polymer coatings form a hydration layer, which is a physical barrier preventing the non-specific adsorption of marine organisms. The hydrophilic yet electrically neutral nature of these polymers also reduces electrostatic interactions with charged biomass.^[7] However, drawbacks of these coatings have recently been identified. When exposed to marine environments, they can lose their antifouling capabilities due to the surface adsorption of minerals present in the sediment. For example, Koc et al. reported that inorganic materials such as diatomite and silt are easily adsorbed on zwitterionic and ethylene glycol polymer coatings.^[8,9] However, amphiphilic copolymer coatings showed

relatively low silt adsorption due to a decrease in surface hydrophilicity.^[10] Moreover, one of the unique advantages of the amphiphilic copolymer coatings is their ability to exhibit both hydrophobic and hydrophilic characteristics at the surface. This property allows them to present an ambiguous surface to biofoulants, making them effective for marine antifouling coatings.^[11]

Various amphiphilic copolymers have been synthesized, and their antifouling properties have been demonstrated.^[10,12–17] For instance, Koschitzki et al. prepared a set of amphiphilic polymers containing carboxybetaine and dicyclopentadienyl, which show good antifouling behavior against marine diatoms.^[10] Polysaccharides have also been used in the preparation of amphiphilic polymer coatings.^[12,13] Hyaluronic acid and chondroitin sulfate were capped with hydrophobic trifluoroethylamine, which significantly inhibited the adhesion of marine bacteria, zoospores, and diatoms.^[12] Recently, nitroxide-containing copolymers were suggested for amphiphilic copolymer coatings.^[15] Several hard-fouling organisms use radical-based crosslinked adhesives; therefore, nitroxides acting as antioxidants when incorporated

1. Introduction

Control of marine biofouling, the accumulation of biomass from various marine organisms on the ship hull, is an ongoing challenge in the maritime industry.^[1,2] Marine biofouling increases surface friction, causing ships to use more fuel than otherwise required.^[3] Moreover, biofouling by alien species results in their

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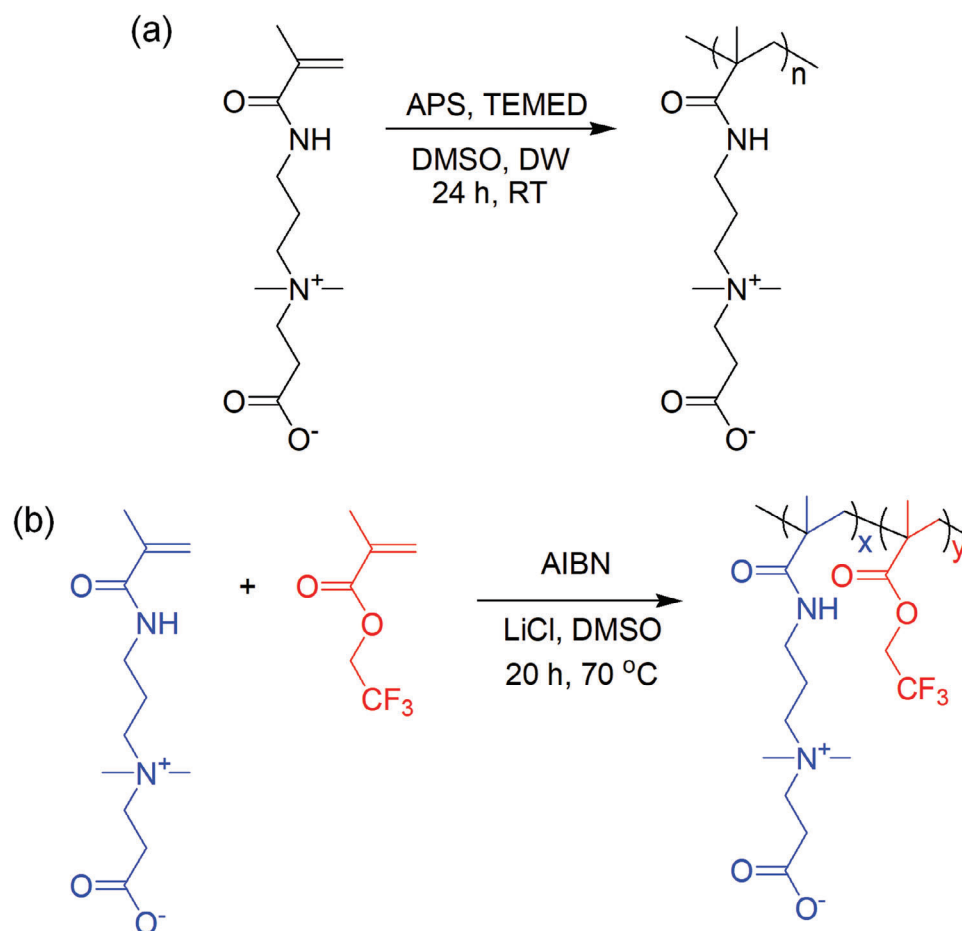


Figure 1. Synthetic schemes for a) poly(CBMA) and b) poly(CBMA-*r*-TFEMA).

into the copolymers can lead to the polymers completely inhibiting barnacle settlement.

A pressing challenge in the fabrication of amphiphilic copolymer coatings is their lack of versatility. Many studies have been conducted on glass or silicon surfaces because these surfaces are easily modified via well-known organosilane chemistry. For example, amino or vinyl groups were first introduced on the glass surface using organosilane compounds, and amphiphilic copolymers were covalently grafted onto the functionalized glass surface using a photo-crosslinker, benzophenone.^[17] In another study, polydimethylsiloxane (PDMS)-based coatings were investigated.^[16] PDMS-containing amphiphilic copolymers were synthesized and incorporated into a PDMS matrix to create coatings. Although these approaches were successful in marine antifouling coatings, additional photoreactive groups or matrices needed to be included in the coating process, which can adversely affect the intrinsic properties of the copolymer. Therefore, a method that can be applied to various surfaces without additional anchoring groups or matrices is required.

Herein, we propose a simple and versatile method for fabricating amphiphilic copolymer coatings. This method utilizes the material-independent surface-coating property and metal complex-forming ability of polydopamine (pDA)^[18] to immobilize amphiphilic copolymers onto solid surfaces. Amphiphilic

copolymers composed of carboxybetaine methacrylamide and trifluoroethyl methacrylate were synthesized and physically deposited onto pDA-coated solid surfaces via spin coating. The deposited copolymers were crosslinked with pDA by zirconium(IV) (Zr^{IV})-mediated multiple coordination bond formation,^[19] enabling the robust immobilization of copolymers onto solid surfaces. Using this method, silicon, glass, and stainless steel substrates were successfully coated with amphiphilic copolymers, and the resulting surfaces exhibited good antifouling performance against both diatom adhesion and silt adsorption.

2. Results and Discussion

Poly(carboxybetaine methacrylamide) (poly(CBMA)) and two random poly(carboxybetaine methacrylamide-*r*-trifluoroethyl methacrylate) (poly(CBMA-*r*-TFEMA)) copolymers were synthesized by free-radical solution polymerization (Figure 1). The CBMA-to-TFEMA ratio in poly(CBMA-*r*-TFEMA) was altered by controlling the amount of monomer added during polymerization.

The CBMA-to-TFEMA ratio was determined by nuclear magnetic resonance (NMR) spectroscopy. The integral of the peak at 4.2 ppm (CF_3CH_2-) was compared with the integral of the peak at 2.6 ppm ($-N^+(CH_3)_2$), and as a result, the CBMA-to-TFEMA

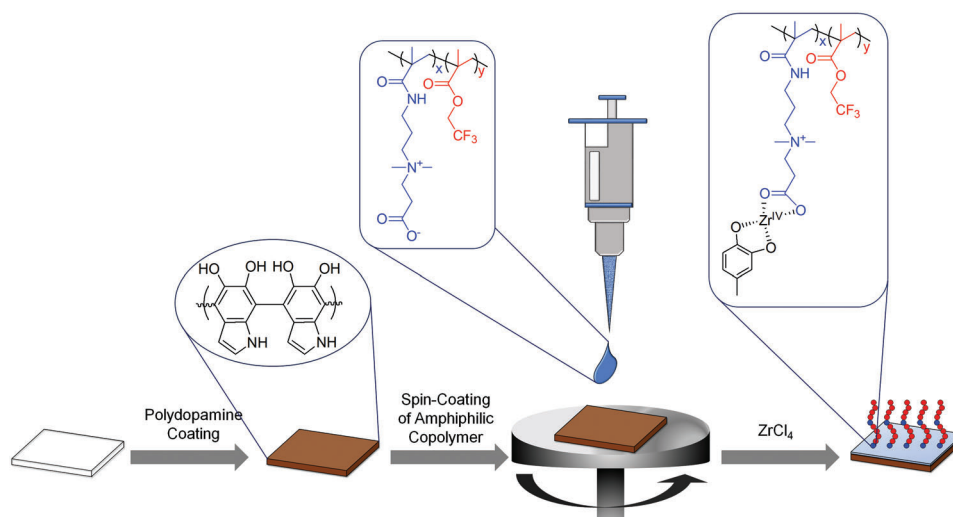


Figure 2. Schematic representation of the pDA-mediated amphiphilic copolymer coating.

ratios of two types of poly(CBMA-*r*-TFEMA) were calculated to be 1:3.3 and 1:6.6, respectively (Figure S1, Supporting Information). These polymers were designated as poly(CBMA₁-*r*-TFEMA_{3.3}) and poly(CBMA₁-*r*-TFEMA_{6.6}). The molecular weights and polydispersities of the synthesized polymers were determined by gel permeation chromatography (GPC) (Table 1; Figure S2, Supporting Information). The M_n values of poly(CBMA), poly(CBMA₁-*r*-TFEMA_{3.3}), and poly(CBMA₁-*r*-TFEMA_{6.6}) were 17.8, 140, and 255 kDa, respectively. The molecular weight distributions of these polymers are 1.68, 2.00, and 2.15.

To immobilize the synthesized polymers onto the solid surface, a pDA coating was performed. pDA coating is a well-known coating method that forms a catechol-presenting solid surface in a material-independent manner.^[18,20] In this study, pDA catechol groups on solid surfaces were used as the anchoring sites for poly(CBMA) and poly(CBMA-*r*-TFEMA). pDA-mediated polymer immobilization was achieved by adding Zr^{IV} ions. Zr^{IV} can form multiple coordination bonds with various organic ligands, such as catechol and carboxylate.^[19] Therefore, it is possible to create catechol-Zr^{IV}-carboxylate coordination bonds between the pDA-coated surface and the CBMA-containing polymer and carboxylate-Zr^{IV}-carboxylate coordination bonds between the CBMA-containing polymers (Figure 2). Si/SiO₂ was initially used as a model substrate for the surface coating because several surface characterization techniques are available for Si/SiO₂ substrates.

Table 1. Molecular weights and polydispersities of the synthesized poly(CBMA) and poly(CBMA-*r*-TFEMA) were determined by GPC calibrated with polyethylene glycol and polymethylmethacrylate standards, respectively.

	M_w [kg mol ⁻¹]	M_n [kg mol ⁻¹]	Polydispersity Index (PDI)
poly(CBMA)	29.9	17.8	1.68
poly(CBMA ₁ - <i>r</i> -TFEMA _{3.3})	280	140	2.00
poly(CBMA ₁ - <i>r</i> -TFEMA _{6.6})	549	255	2.15

The pDA surface coating and subsequent polymer immobilization were confirmed by spectroscopic ellipsometry (Figure 3a). The pDA coating thickness was measured to be 7.0 ± 0.1 nm, and the thickness increased by 8.1 ± 0.2 , 41.6 ± 3.3 , and 42.5 ± 2.9 nm, after immobilizing of poly(CBMA), poly(CBMA₁-*r*-TFEMA_{3.3}), and poly(CBMA₁-*r*-TFEMA_{6.6}), respectively. These results show that the immobilization efficiency of the copolymers was higher than that of poly(CBMA). This might be attributed to the difference in molecular weights. Because the carboxylate acts as a surface-anchoring group, the higher the molecular weight, the greater the number of carboxylates participating in polymer immobilization, thereby increasing the immobilization efficiency.^[21] The surface wettability following polymer immobilization was investigated using water contact angle (WCA) goniometry (Figure 3b). Although there was a difference in polymer immobilization efficiency, as evidenced by spectroscopic ellipsometry, the surface wettability was significantly altered by polymer immobilization. The WCAs of the uncoated Si/SiO₂ surface and pDA coating were $45.9^\circ \pm 1.0^\circ$ and $35.2^\circ \pm 1.0^\circ$. After immobilization with poly(CBMA), the WCA decreased to $<10^\circ$ because of the strong binding affinity of CBMA for water molecules.^[22] In contrast, the WCA increased to $88.0^\circ \pm 0.3^\circ$ and $93.0^\circ \pm 7.4^\circ$ when coated with poly(CBMA₁-*r*-TFEMA_{3.3}) and poly(CBMA₁-*r*-TFEMA_{6.6}), respectively. These results indicate that TFEMA was mainly exposed on the outermost poly(CBMA-*r*-TFEMA) layer under dry conditions, making the surface relatively hydrophobic.

Our approach for fabricating amphiphilic copolymer coatings has the advantage of being applicable to various solid substrates owing to the material-independent surface coating properties of pDA. To investigate its versatility, amphiphilic copolymer immobilization was performed on stainless steel (SS) substrates because SS is the major element of the ship hull, on which marine fouling frequently occurs. First, the changes in surface wettability were investigated. Similar to coatings on Si/SiO₂ surfaces, the WCA of SS was changed to $36.6^\circ \pm 1.1^\circ$ from $65.7^\circ \pm 5.8^\circ$, after pDA coating. After immobilization of poly(CBMA), the WCA further decreased to $<10^\circ$, and

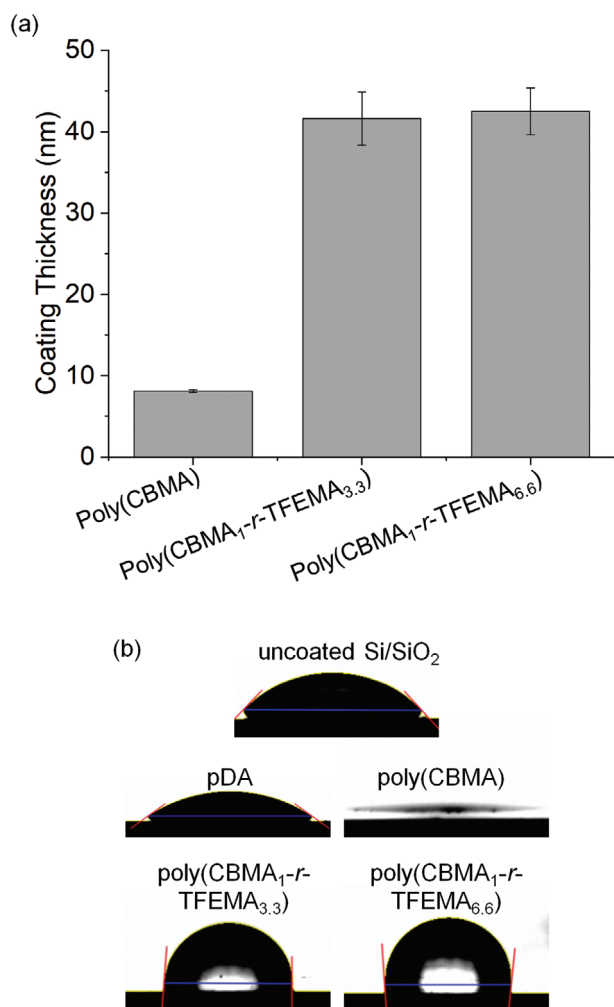


Figure 3. a) Thicknesses of poly(CBMA), poly(CBMA₁-r-TFEMA_{3.3}), and poly(CBMA₁-r-TFEMA_{6.6}) coatings on Si/SiO₂. b) WCA images of uncoated Si/SiO₂, pDA, poly(CBMA), poly(CBMA₁-r-TFEMA_{3.3}), and poly(CBMA₁-r-TFEMA_{6.6}) coatings.

immobilization of poly(CBMA₁-r-TFEMA_{3.3}) and poly(CBMA₁-r-TFEMA_{6.6}) made the surface hydrophobic: $79.4^\circ \pm 2.2^\circ$ for poly(CBMA₁-r-TFEMA_{3.3}) and $104.7^\circ \pm 5.5^\circ$ for poly(CBMA₁-r-TFEMA_{6.6}) (Figure 4a).

Successful polymer immobilization was confirmed by X-ray photoelectron spectroscopy (XPS) (Figure 4b). The XPS spectrum of the uncoated SS surface displayed three strong substrate peaks for C 1s, O 1s, and Fe 2p. After the pDA coating, the substrate peaks for Fe 2p disappeared, and an increase in the peak intensity for C 1s and N 1s was observed. These results indicated that the pDA coating on the SS was sufficiently thick such that substrate peaks were not detected. The polymer immobilization onto the pDA coating further changed the surface chemical composition. In the spectrum of the poly(CBMA)-immobilized surface, the peaks for Zr 3d and quaternary ammonium appeared (Figure 4b; Figure S3, Supporting Information). The poly(CBMA-r-TFEMA) copolymer immobilization was also characterized by the appearance of Zr 3d and F 1s peaks. In addition, we used surface atomic composition data to analyze the proportions of

CBMA and TFEMA exposed on the surface (Table 2). As the thicknesses of the copolymers were greater than the detection limit of XPS, we assumed that nitrogen and fluorine originated only from CBMA and TFEMA and not from pDA. Considering that the CBMA and TFEMA monomer units contain two nitrogen atoms and three fluorine atoms, respectively, the ratios of CBMA and TFEMA exposed to the surface in the poly(CBMA₁-r-TFEMA_{3.3}) and poly(CBMA₁-r-TFEMA_{6.6}) coatings were calculated to be 3.1 and 6.4. These results are in good agreement with the composition ratios of the copolymers obtained from the NMR analyses (3.3 and 6.6).

The surface morphologies and root-mean-square (RMS) roughnesses of the polymer were determined by atomic force microscopy (AFM) (Figure 5). Uncoated SS and pDA coating displayed similar surface morphology, with the pDA coating increasing the RMS roughness (from 7.6 nm to 14.7 nm). Polymer immobilization onto the pDA-coated surface significantly changed the surface morphology; many protrusions appeared on all polymer-immobilized surfaces, whereas only a few protrusions were found on the uncoated and pDA-coated SS surfaces. The RMS roughness values of the poly(CBMA), poly(CBMA₁-r-TFEMA_{3.3}), and poly(CBMA₁-r-TFEMA_{6.6}) coatings were 26.4, 43.6, and 15.9 nm, respectively.

In order to investigate the polymer coating stability in an ionic solution, coating samples were incubated in an artificial seawater for two weeks. The coating thicknesses were analyzed by spectroscopic ellipsometry before and after seawater treatment. Figure S4 (Supporting Information) shows that at least 80% of the coating thickness was retained upon the exposure to seawater for two weeks. Specifically, poly(CBMA), poly(CBMA₁-r-TFEMA_{3.3}), and poly(CBMA₁-r-TFEMA_{6.6}) coatings lost 20%, 2%, and 9% of their coating thickness, respectively, implying the robustness of Zr^{IV}-mediated coordination bonds under high-salt condition.

The adhesion behavior of marine diatoms was studied to investigate the marine antifouling properties of the polymer coatings. Marine diatom adhesion assays were conducted under both static and dynamic conditions. Under static conditions, the polymer-immobilized surfaces were immersed in a marine diatom suspension without shaking, allowing marine diatoms to access and adhere to the solid surfaces by gravity. Surface wettability is a major factor determining the adhesion behavior of marine diatoms.^[5,23] It was expected that the more hydrophilic the surface, the better the antifouling performance because the hydration layers along the hydrophilic polymer serve as a physical barrier to inhibit marine diatom adhesion. Figure 6a shows that poly(CBMA) immobilization suppressed marine diatom

Table 2. Atomic composition (%) of uncoated SS, pDA, poly(CBMA), poly(CBMA₁-r-TFEMA_{3.3}), and poly(CBMA₁-r-TFEMA_{6.6}) coatings.

Atomic %	C 1s	F 1s	O 1s	N 1s	Zr 3d	Fe 2p
uncoated SS	29.51	0.36	47.77	1.61	0.1	17.63
pDA	70.87	0.39	19.25	8.66	0.07	0.27
poly(CBMA)	54.07	0.55	33.55	6.43	2.79	1.5
poly(CBMA ₁ -r-TFEMA _{3.3})	57.47	16.31	19.37	3.48	1.04	1.19
poly(CBMA ₁ -r-TFEMA _{6.6})	53.88	21.63	18.52	2.25	0.65	1.57

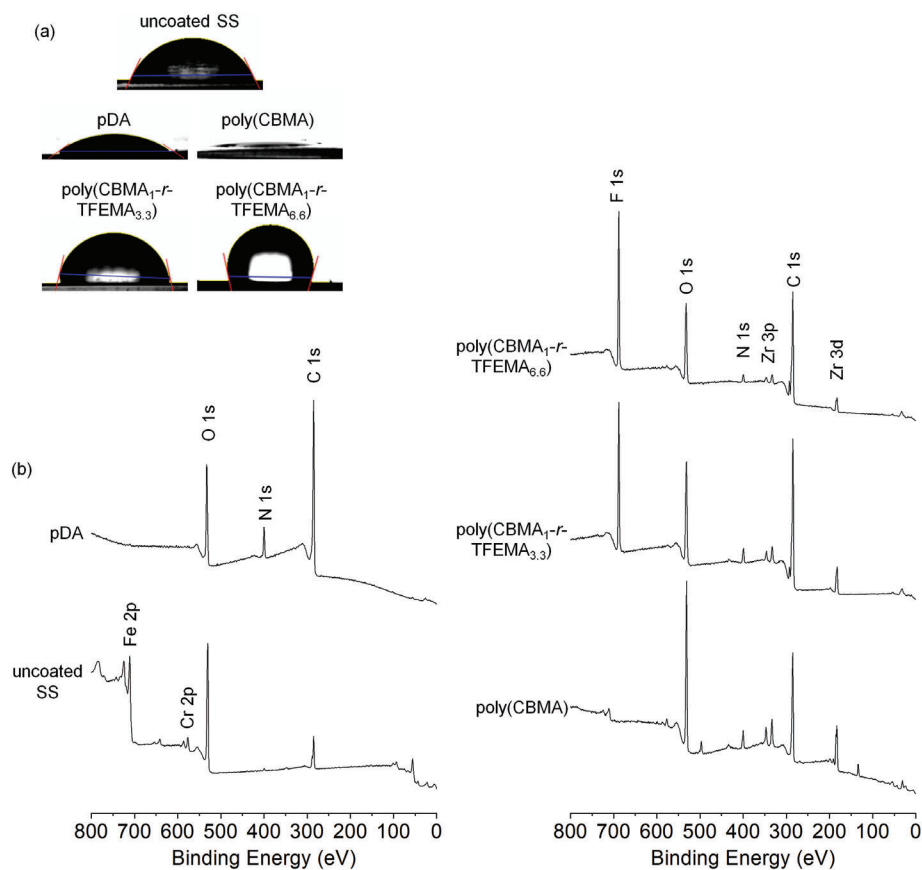


Figure 4. a) WCA images and b) XPS spectra of uncoated SS, pDA, poly(CBMA), poly(CBMA₁-*r*-TFEMA_{3,3}), and poly(CBMA₁-*r*-TFEMA_{6,6}) coatings.

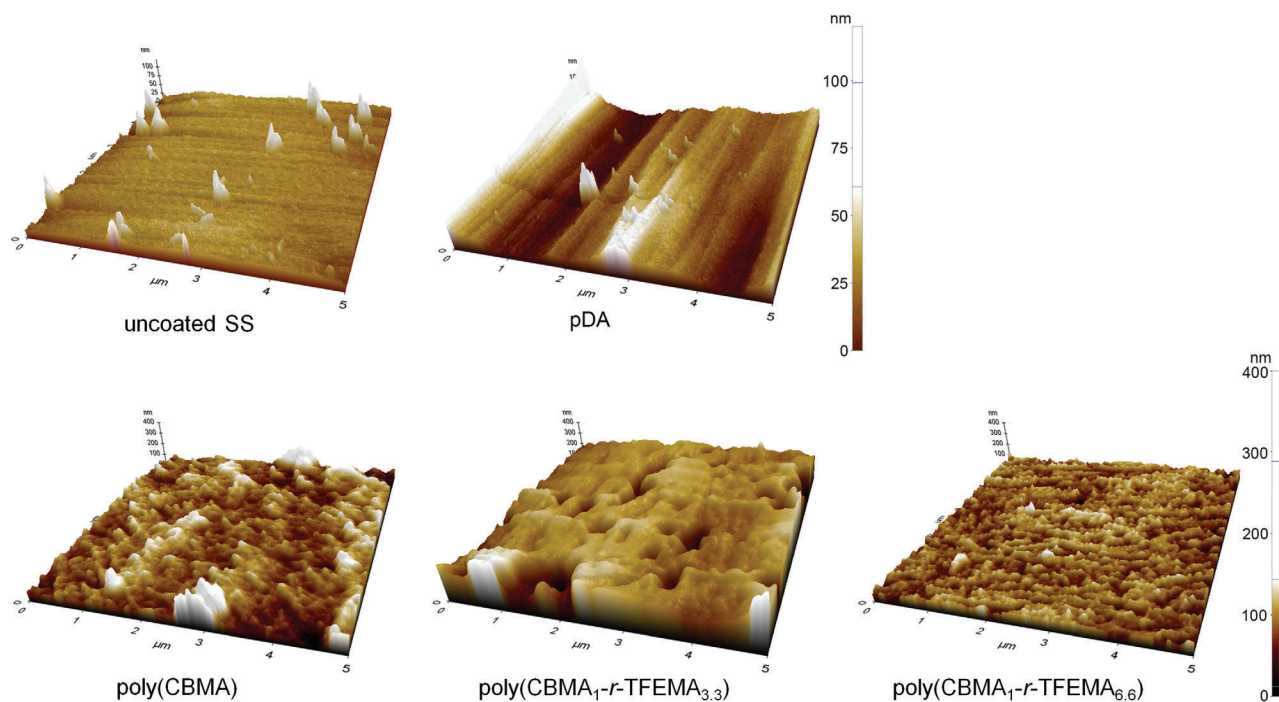


Figure 5. AFM images of uncoated SS, pDA, poly(CBMA), poly(CBMA₁-*r*-TFEMA_{3,3}), and poly(CBMA₁-*r*-TFEMA_{6,6}) coatings.

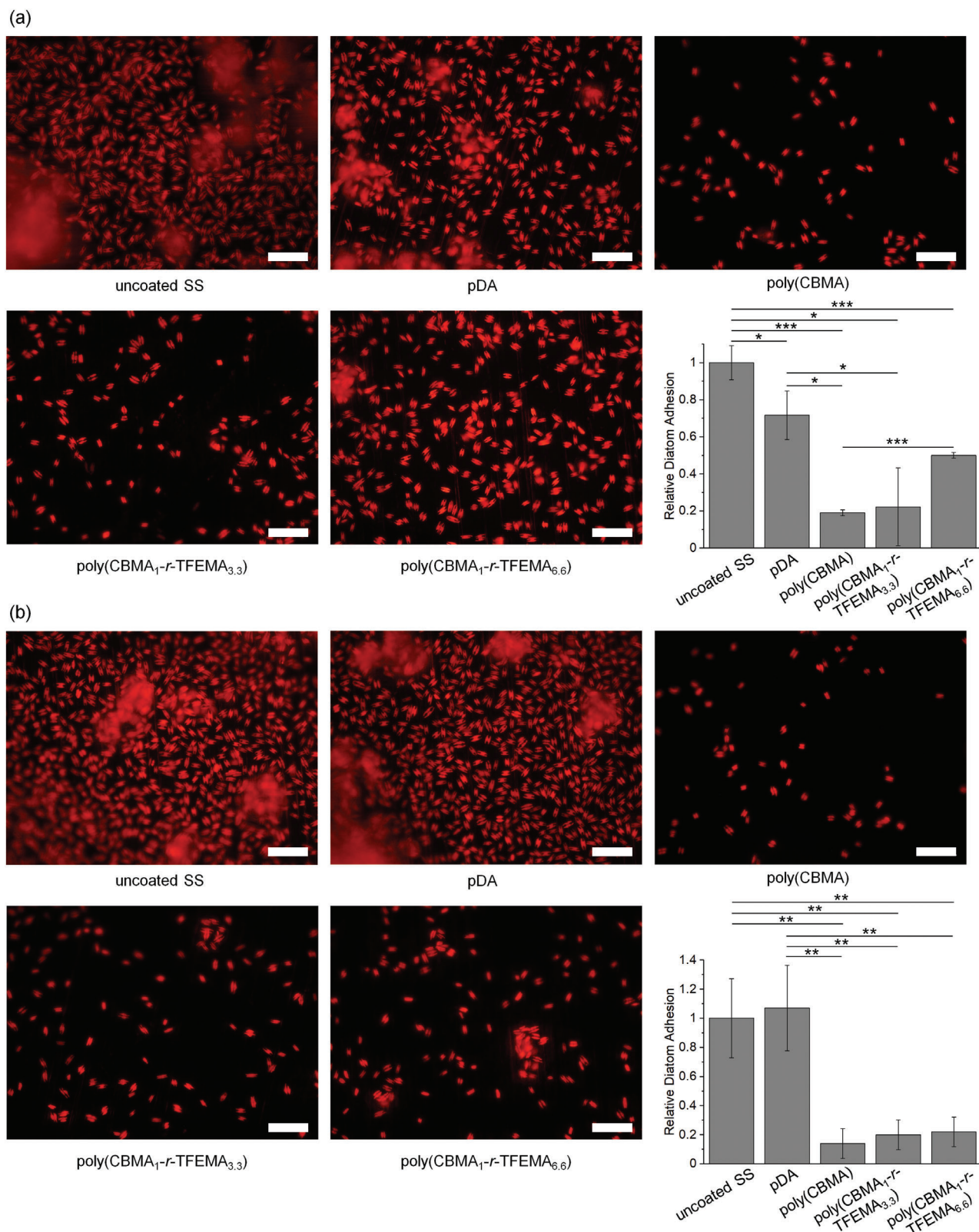


Figure 6. Fluorescence images and quantification of diatom adhesion on the uncoated SS, pDA, poly(CBMA), poly(CBMA_{1-r}-TFEMA_{3,3}), and poly(CBMA_{1-r}-TFEMA_{6,6}) coatings under a) static and b) dynamic conditions. Each value is the mean of 60 counts of three replicate samples ($n = 3$). The error bars represent the 95% confidence limit. The scale bars represent 50 μm . Significant differences between the two groups were analyzed via unpaired Student's *t*-test. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

adhesion by 81% compared to diatom adhesion on the uncoated SS surfaces. Interestingly, despite increased surface hydrophobicity, no significant increase in diatom adhesion was observed on the poly(CBMA₁-*r*-TFEMA_{3.3})-immobilized surfaces. In contrast, the attached diatom density on the poly(CBMA₁-*r*-TFEMA_{6.6})-immobilized surfaces was higher than that on the poly(CBMA)-immobilized surfaces. Quantitatively, diatom adhesion was only reduced by 50% on the poly(CBMA₁-*r*-TFEMA_{6.6})-immobilized surfaces compared to diatom adhesion on the uncoated SS surfaces. These results indicate that adding hydrophobic monomers (i.e., TFEMA) to zwitterionic marine antifouling polymers can cause adverse effects if used under static conditions.

Under dynamic conditions, polymer-immobilized surfaces were immersed in a marine diatom suspension with continuous shaking. In such a circumstance, the fouling-release property originating from the coating with low-surface-energy materials also affects the adhesion behavior of marine diatoms. As shown in Figure 6b, poly(CBMA) immobilization significantly inhibited marine diatom adhesion, reducing it by 86% under dynamic conditions, and a similar property was also observed on the poly(CBMA₁-*r*-TFEMA_{3.3})- and poly(CBMA₁-*r*-TFEMA_{6.6})-immobilized surfaces. The attached diatom density increased slightly, but not significantly, on the copolymer-immobilized surfaces compared to the poly(CBMA)-immobilized surfaces. This is attributed to the combination of the antifouling properties of CBMA and the fouling-release properties of TFEMA. The zwitterionic CBMA inhibited the initial diatom adhesion while the hydrophobic TFEMA weakened diatom adhesion to the solid surface. Consequently, the copolymer-immobilized surfaces exhibited antifouling properties similar to those of poly(CBMA) immobilization.

For silt adsorption test, pDA coating and subsequent polymer immobilization were performed on glass substrates. The coating samples were immersed in an autoclaved sediment suspension for 60 min, and the amount of silt adsorbed onto the coating was determined by subtracting the absorbance (at 440 nm) before the silt adsorption test from that after the test (Figure 7), as the absorbance increased when the silt was adsorbed.^[8] Silt adsorption occurred on all the polymer-immobilized surfaces. The highest silt adsorption was observed on poly(CBMA)-immobilized surfaces, whereas the lowest silt adsorption was observed on poly(CBMA₁-*r*-TFEMA_{6.6})-immobilized surfaces. The adsorption of silt on poly(CBMA₁-*r*-TFEMA_{3.3})-immobilized surfaces was higher than that on poly(CBMA₁-*r*-TFEMA_{6.6})-immobilized surfaces. These results confirm the critical role of TFEMA in poly(CBMA-*r*-TFEMA) in reducing silt adsorption. Furthermore, more than a polymer coating composed solely of CBMA is required for use in the field.

3. Conclusion

In summary, we synthesized amphiphilic copolymers composed of carboxybetaine- and fluorine-containing monomers and developed a facile method for effectively immobilizing the copolymers onto solid substrates for marine antifouling applications. This method, mediated by mussel-inspired polydopamine, allowed for the successful immobilization of copolymers onto

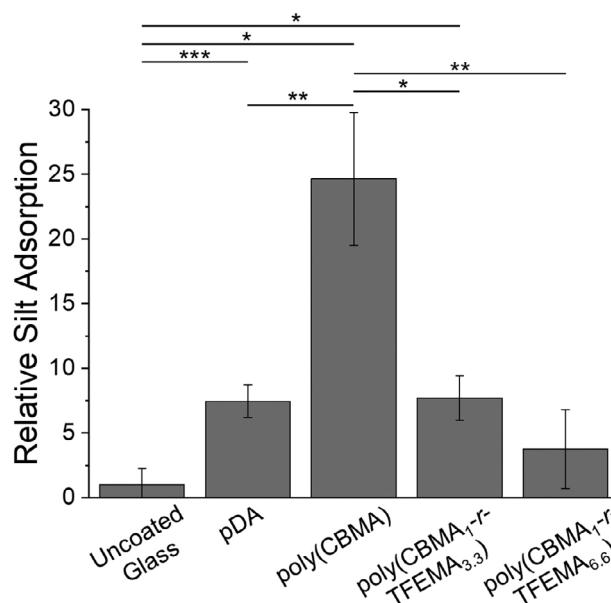


Figure 7. Silt adsorption on the polymer coatings after 60-min sand immersion. The absorbance of the coatings at a wavelength of 440 nm was measured before and after the sand immersion (Substrate: glass). Each point is the 9-count mean on three replicated samples ($n = 3$). The error bars represent the 95% confidence limit. The amount of adsorbed silt on the uncoated glass surface was considered 1. Significant differences between the two groups were analyzed via unpaired Student's *t*-test. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

stainless steel surfaces. Diatom adhesion and silt adsorption tests were conducted to verify the antifouling performance of the copolymer-immobilized surfaces, and the experimental results were compared with those obtained using poly(carboxybetaine)-immobilized surfaces. The results revealed that the antifouling performance of poly(carboxybetaine) immobilization was the best in the diatom adhesion assay under static conditions and was similar to that of copolymer immobilization under dynamic conditions. However, poly(carboxybetaine) immobilization showed much better silt adsorption behavior than copolymer immobilization; therefore, it is expected to have a poorer antifouling performance than copolymer immobilization in field applications. As the method investigated herein can be applied to diverse surfaces and allows the immobilization of various polymers, we expect it to be an effective platform for the coating of new marine antifouling polymers.

4. Experimental Section

Materials: Dopamine hydrochloride (DA, 98%; Sigma–Aldrich), tris(hydroxymethyl) aminomethane hydrochloride (Trizma HCl, 99%; Sigma–Aldrich), and tris(hydroxymethyl) aminomethane (Trizma base, 99%; Sigma–Aldrich), *N,N,N',N'*-tetramethylethylenediamine (TEMED, 99%; Sigma–Aldrich), ammonium persulfate (APS, 98%; Sigma–Aldrich), dimethyl sulfoxide (DMSO, 99%; Sigma–Aldrich), phosphate-buffered saline (PBS; Sigma–Aldrich), *N*-(3-dimethylaminopropyl)-methacrylamide (99%; Sigma–Aldrich), lithium chloride (LiCl, > 99.0%, Sigma–Aldrich), 2,2,2-trifluoroethyl methacrylate (TFEMA, 99%, Sigma–Aldrich), anhydrous tetrahydrofuran (THF, 99.9%; Samchun), β -propiolactone (> 95%; TCI), 2,2'-azobis(2-methylpropionitrile) (AIBN, 98%, Junsei), and

zirconium(IV) chloride (ZrCl_4 ; Alfa Aesar) were used as received. The marine benthic diatom *Amphora coffeaeformis* was obtained from the Korea Marine Microalgae Culture Center (KMMCC) and grown in 100 mL of f/2 culture medium at 18 °C for 10 days.

Synthesis of (3-methacryloylaminoethyl)-(2-carboxy-ethyl)-dimethylammonium (CBMA): CBMA was synthesized by referring to a previous study.^[24] N-(3-dimethylaminopropyl)-methacrylamide (4.8 mL, 26.8 mmol) was dissolved in 27 mL of anhydrous THF and cooled to 0 °C. β -Propiolactone (2.1 mL, 33.5 mmol) was separately dissolved in anhydrous THF (44 mL) and added dropwise under a nitrogen atmosphere for 2–3 h. After the 30 h reaction at 0–4 °C, the product was obtained as a white precipitate, which was then filtered off and washed with THF and ether. Finally, the product was dried under high vacuum.

Synthesis of Poly(CBMA) and its Coating on Solid Substrates: CBMA (1.2 g, 4.95 mmol) was dissolved in a co-solvent consisting of DMSO (3 mL) and deionized (DI) water (15 mL). Nitrogen gas was bubbled into the solution for 1 h to remove oxygen. Subsequently, APS (23.6 mg) and TEMED (20 μL) were added to the solution, and polymerization was performed at room temperature under a nitrogen gas atmosphere for 25 h. The product was dialyzed (molecular weight cut-off [MWCO] = 3500) against DI water for 3 d and then freeze-dried for 3–5 days.

Prior to the poly(CBMA) coating, the solid substrates were cleaned with ethanol under sonication for 5 min and dried under a stream of nitrogen gas. The solid substrates were then immersed in a DA solution (10 mM, Tris buffer, pH 8.4) for 3 h and washed with DI water. Poly(CBMA) was dissolved in 1 $\times$ PBS (10 mg mL⁻¹, pH 7.4) and dropped onto the pDA-coated substrates, which were then rotated at 3000 rpm for 180 s. Poly(CBMA)-immobilized substrates were immediately immersed in a methanolic solution of ZrCl_4 (0.1 M) for 10 min, rinsed with DI water, and dried under a stream of nitrogen gas. The substrates were then immersed in 1 $\times$ PBS (pH 7.4) for 1 h, rinsed with DI water, and dried under a stream of nitrogen gas.

Synthesis of Poly(CBMA-r-TFEMA) and Its Coating on Solid Substrates: To synthesize poly(CBMA₁-r-TFEMA_{3,3}), CBMA (2 g), TFEMA (3 g), and AIBN (5 mg) were dissolved in DMSO (40 mL) containing 0.05 g of LiCl. Nitrogen gas was bubbled into the solution for 30 min to remove oxygen, and polymerization was then carried out at 70 °C for 20 h under a nitrogen gas atmosphere. The copolymer product was precipitated using a 1:1 mixture of ethanol and hexane. The product was vacuum filtered and stirred in methanol for 24 h to remove the unreacted reagents. The final product was vacuum filtered and dried under high vacuum conditions. The synthesis of poly(CBMA₁-r-TFEMA_{6,6}) was carried out in the same manner as poly(CBMA₁-r-TFEMA_{3,3}), except that 6 g of TFEMA was used. Prior to poly(CBMA-r-TFEMA) coating, the solid substrates were cleaned with ethanol under sonication for 5 min and dried under a stream of nitrogen gas. The solid substrates were then immersed in a DA solution (10 mM, Tris buffer, pH 8.4) for 3 h and washed with DI water. Poly(CBMA-r-TFEMA) was dissolved in trifluoroethanol (10 mg mL⁻¹) and dropped onto the pDA-coated substrates, which were then rotated at 3000 rpm for 180 s. The polymer-immobilized substrates were immediately immersed in a 1:1 mixture of methanol and DI water containing ZrCl_4 (0.1 M) for 10 min, rinsed with DI water, and dried under a stream of nitrogen gas. The substrates were then immersed in trifluoroethanol for 10 min, rinsed with DI water, and dried under a stream of nitrogen gas.

Characterization: *Material characterization:* The molecular weight of the poly(CBMA) was determined by GPC using RID-20 A (Shimadzu Corp., Japan). The GPC column (OHpak SB-804 HQ, Shodex, Munich, Germany) was protected using a guard column (OHpak SB-G 6 B, Shodex). An aqueous solution of sodium nitrate (0.1 M, pH 7.4) was used as the eluent, and the flow rate was set to 0.5 mL min⁻¹ at 30 °C. The molecular weight of poly(CBMA-r-TFEMA) was determined by GPC using an RI detector (Tosoh EcoSEC HLC-8320, Japan). The GPC column (TSKgel SuperAWM-Hx2) was protected with a guard column (TSKgel guard column SuperAW-H). A hexafluoroisopropanol solution of sodium trifluoroacetic acid (10 mM) was used as the eluent at a flow rate set to 0.3 mL min⁻¹ at 40 °C. NMR spectroscopy (AVANCE 500 MHz, Bruker) was used to determine the poly(CBMA) to poly(TFEMA) ratio.

Coated substrate characterization: XPS was conducted using a K-Alpha⁺ spectrometer (Thermo Scientific, UK) equipped with an Al K α X-ray source. The thickness of the organic coat on the solid substrates was measured using an alpha-SE spectroscopic ellipsometer (J. A. Woollam) in the wavelength range of 390–900 nm at an incidence angle of 70°. The data were fitted to a Cauchy model using CompleteEASE software (refractive index $n(\lambda) = 1.45 + 0.01/\lambda^2$). Static WCA measurements (Phoenix-10, Surface Electro Optics, Suwon, South Korea) were used to determine the hydrophobicity of the coatings. AFM images were obtained in the non-contact mode using an XE-7 microscope (Park Systems, Suwon, South Korea). Finally, fluorescence images were obtained using an optical microscope equipped with a fluorescence imaging system (LV100ND; Nikon, Tokyo, Japan).

Static Marine Diatom Adhesion Test: A suspension of marine diatoms with a 3 $\mu\text{g mL}^{-1}$ chlorophyll content was prepared. The diatom suspension was spread onto the solid substrates and incubated at 20 °C for 24 h without shaking. The solid substrates were gently washed with artificial seawater. The fouling behavior of diatoms on the solid substrates was analyzed using fluorescence microscopy.

Dynamic Marine Diatom Adhesion Test: The diatom suspension was spread onto a solid substrate and incubated in a reciprocating shaker (MaXshakeRP2; Daihan Scientific Co.) for 24 h at 50 rpm. The solid substrates were gently washed with artificial seawater. The fouling behavior of diatoms on the solid substrates was analyzed using fluorescence microscopy.

Silt Adsorption Test: Natural sand sediment from the test site at Port Boryeong, South Korea (36°20'15.1"N 126°31'57.5" E) was autoclaved and suspended in sterile filtered seawater at a concentration of 37.5 g L⁻¹. Glass substrates were used for the silt adsorption test, and both sides (top and bottom) of the substrates were immobilized with polymers. The solid substrates were then immersed in the sediment suspension and incubated in a reciprocating shaker for 60 min at 50 rpm. After gently washing with artificial seawater, the solid substrates were dried under ambient conditions. On each sample surface, the photometric absorption at 440 nm (UV-1800, Shimadzu, Tokyo, Japan) was measured and compared with the absorbance of the non-silt-adsorbed substrate.

Statistical Analysis: Three independent samples were prepared. The data are presented as the mean $\pm$ 95% confidence limit. Statistical significance was determined using an unpaired, two-sided Student's *t*-test ($n = 3$). *** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

amphiphilic polymer, marine antifouling, metal complexation, polydopamine, surface coating

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